

W03

Algebra for Honor Mathematics

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§1 Q1

(a)

$$\begin{aligned}\overline{G} &= \{3, 5, 7, 8, 10\} \\ \overline{G} \cap H &= \{5, 7\}.\end{aligned}$$

(b)

$$\begin{aligned}F \cup H &= \{1, 4, 5, 7, 9, 10\} \\ G \cap (F \cup H) &= \{1, 4, 9\}\end{aligned}$$

(c)

$$\begin{aligned}G \cup H &= \{1, 2, 4, 5, 6, 7, 9\} \\ \overline{(G \cup H)} &= \{3, 8\}\end{aligned}$$

(d)

$$\begin{aligned}(F \cap H) &= \{4, 5, 9\} \\ (F \cap H) - G &= \{5\}\end{aligned}$$

§2 Q2

Prove the following statement.

$$\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, (x^2 + 2 - y) \mid (10 - x^2 - y^2)$$

Solution. Let x be arbitrary number in $\mathbb{N}$. We select y to be 3 in $\mathbb{N}$. substituting $y = 3$ into the expression, we

$$(x^2 + 2 - 3) \mid (10 - x^2 - (3)^2)$$

This implies

$$(x^2 - 1) \mid (10 - x^2 - 9),$$

which equals to

$$(x^2 - 1) \mid (1 - x^2)$$

there exists a number $k = -1$ in $\mathbb{Z}$ such that

$$k(x^2 - 1) = -1(x^2 - 1) = -x^2 + 1 = 1 - x^2.$$

for all $x \in \mathbb{N}$.

Thus, according to the definition of divisibility,

$$\forall x \in \mathbb{N}, \exists y \in \mathbb{N}, (x^2 + 2x - y) \mid (10 - x^2 - y^2).$$

■

§3 Q3

Prove the following statement.

Solution. Let p, q, r be arbitrary in $\mathbb{Z}$.

By Elimination, it suffices to prove that

$$[(p^2q \mid (r^2 - q)) \wedge (pq^3 \mid qr)] \implies (pq \mid (q(7 + 5r) - 7r^2))$$

Suppose $(p^2q \mid (r^2 - q))$ and $(pq^3 \mid qr)$, by the definition of divisibility, there exists $a, b \in \mathbb{Z}$ such that

$$\begin{aligned} r^2 - q &= p^2qa \\ qr &= pq^3b. \end{aligned}$$

Moreover, we expand $q(7 + 5r) - 7r^2$, we get

$$7q + 5qr - 7r^2.$$

Rerarrange and factoring -7 out of $-7r^2 + 7q$ we obtains

$$-7r^2 + 7q + 5qr = -7(r^2 - q) + 5qr.$$

Subtitute $r^2 - q = p^2qa$ and $qr = pq^3b$ in, we get

$$-7p^2qa + 5pq^3b.$$

Factoring pq out, we get

$$pq(-7pa + 5q^2b).$$

Since $p, q, r, a, b \in \mathbb{Z}$, therefore

$$-7pa + 5q^2b \in \mathbb{Z}$$

such that

$$pq(-7pa + 5q^2b) = q(7 + 5r) - 7r^2.$$

Thus, by the definition of divisibility, we obtain

$$pq \mid (q(7 + 5r) - 7r^2.)$$

■

§4 Q4

(a) **Solution.** Suppose x be arbitrary in $\mathcal{F}$. By the definition of set, we have

$$42 \mid x + 18, x \in \mathbb{Z}.$$

By the definition of divisibility, there exists $k \in \mathbb{Z}$ such that

$$42k = x + 18.$$

Rearrange, we get

$$x = 42k - 18.$$

Factor 3 out, we get

$$x = 3(14k - 6).$$

Since $k \in \mathbb{Z}$,

$$x = 3(14k - 6), 14k - 6 \in \mathbb{Z}.$$

This yields $x \in \mathcal{K}$.

Moreover, substitute $x = 42k - 18$ into $x + 18$, we get

$$42k - 18 + 18 = 42k.$$

This is equal to

$$7 \cdot (6k).$$

Since $k \in \mathbb{Z}$, $6k \in \mathbb{Z}$.

Therefore, there exists $6k \in \mathbb{Z}$ such that

$$x + 18 = 7 \cdot 6k.$$

By the definition of divisibility, we have

$$7 \mid x + 18.$$

We obtain $x \in \mathcal{P}$.

Therefore, we proved

$$\forall x \in \mathbb{Z}, [(x \in \mathcal{F}) \implies (x \in (\mathcal{K} \cap \mathcal{P}))]$$

■

- (b) **Solution.** We disprove this statement by giving a counterexample. Let $x = 45 \in \mathbb{Z}$.
 Since

$$45 + 18 = 63$$

and

$$42 \nmid 63.$$

We have

$$45 \notin \mathcal{F}$$

However, $3 \mid 45$. So

$$45 \in \mathcal{K}.$$

Moreover,

$$45 + 18 = 63.$$

and

$$7 \mid 63.$$

So

$$45 \in \mathcal{P}.$$

Thus, we proved that

$$\exists x \in \mathbb{Z}, [(x \notin \mathcal{F}) \implies (x \in (\mathcal{K} \cap \mathcal{P}))]$$

So, we disprove the original statement. ■